Experiment No. 1

FLASHING OF LED USING AT89C51 MICROCONTROLLER USING PROTEUS

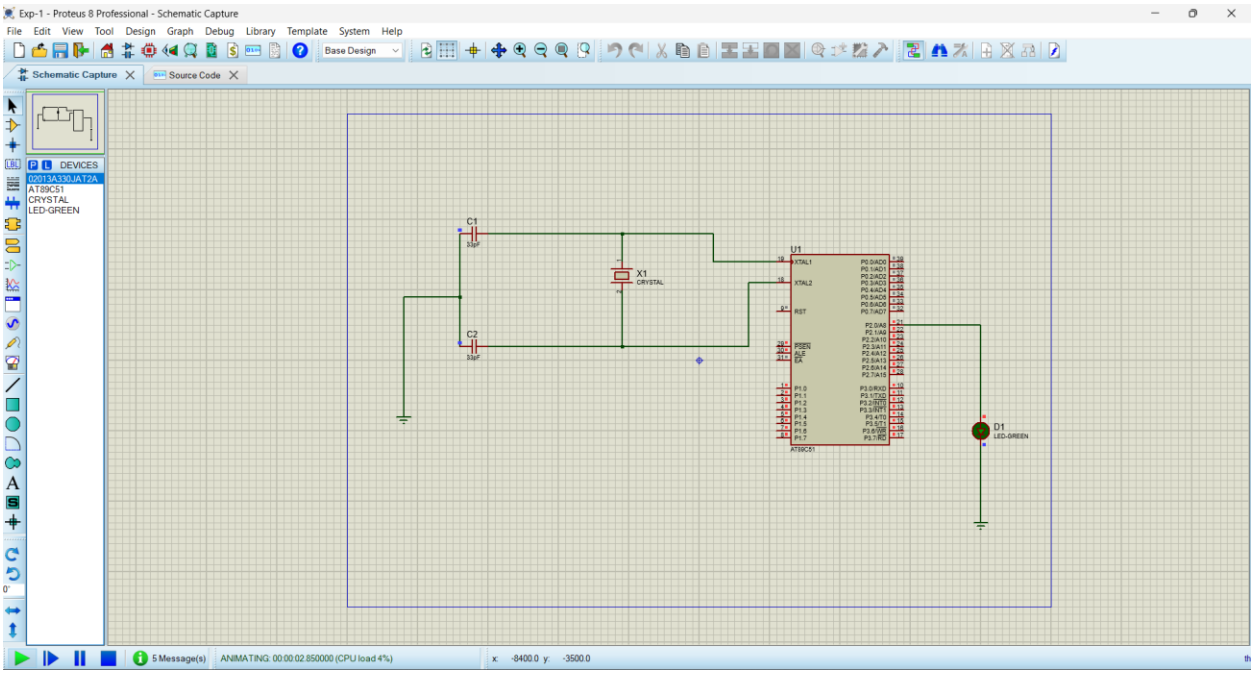
Objectives

1. To study the basic operation of AT89C51 microcontroller.
2. To interface an LED with microcontroller.
3. To generate LED blinking using a software delay.

Program:

```
ORG 0000H
UP: SETB P2.0
ACALL DELAY
CLR P2.0
ACALL DELAY
SJMP UP
DELAY: MOV R4,#35
H1:MOV R3,#255
H2:DJNZ R3,H2
DJNZ R4,H1
RET
END
```

Output:


Exp-1 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Base Design
Schematic Capture
Source Code
DEVICES
02013A330JAT2A
AT89C51
CRYSTAL
LED-GREEN
C1
33pF
C2
33pF
X1
CRYSTAL
U1
XTAL1
XTAL2
RST
PSEN
ALE
EA
AT89C51
D1
LED-GREEN
5 Message(s)
ANIMATING: 00:00:02.850000 (CPU load 4%)
x: -8400.0 y: -3500.0